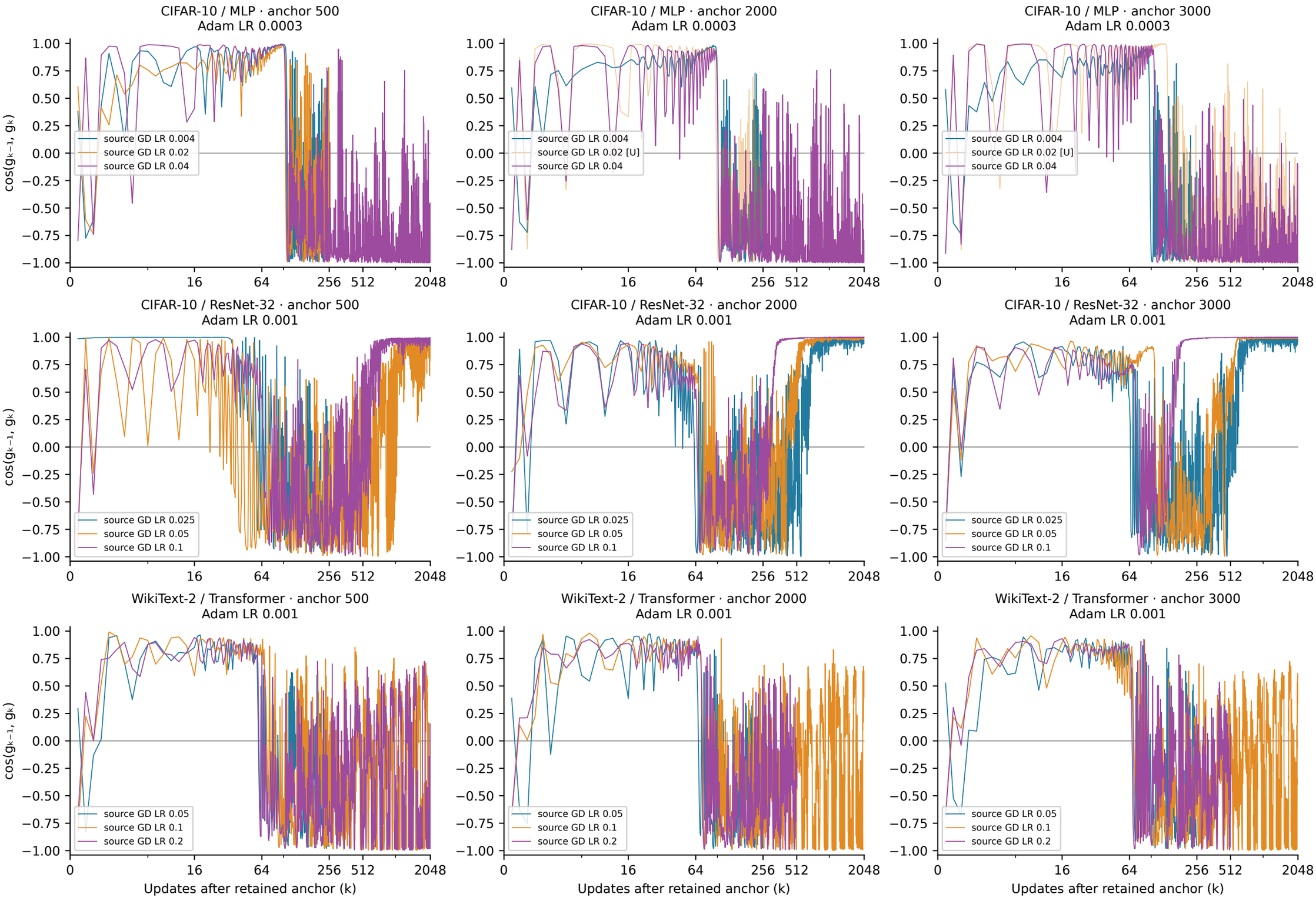

Exact adjacent raw gradients along Adam: matched GD horizons
CIFAR-10 / MLP · anchor 500
Adam LR 0.0003
CIFAR-10 / MLP · anchor 2000
Adam LR 0.0003
CIFAR-10 / MLP · anchor 3000
Adam LR 0.0003
CIFAR-10 / ResNet-32 · anchor 500
Adam LR 0.001
CIFAR-10 / ResNet-32 · anchor 2000
Adam LR 0.001
CIFAR-10 / ResNet-32 · anchor 3000
Adam LR 0.001
WikiText-2 / Transformer · anchor 500
Adam LR 0.001
WikiText-2 / Transformer · anchor 2000
Adam LR 0.001
WikiText-2 / Transformer · anchor 3000
Adam LR 0.001
cos(gₖ₋₁, gₖ)
Updates after retained anchor (k)
source GD LR 0.004
source GD LR 0.02
source GD LR 0.04
source GD LR 0.004
source GD LR 0.02 [U]
source GD LR 0.04
source GD LR 0.004
source GD LR 0.02 [U]
source GD LR 0.04
source GD LR 0.025
source GD LR 0.05
source GD LR 0.1
source GD LR 0.025
source GD LR 0.05
source GD LR 0.1
source GD LR 0.025
source GD LR 0.05
source GD LR 0.1
source GD LR 0.05
source GD LR 0.1
source GD LR 0.2
source GD LR 0.05
source GD LR 0.1
source GD LR 0.2
source GD LR 0.05
source GD LR 0.1
source GD LR 0.2
[U]: unqualified historical reconstruction, excluded from primary summaries and LR selection. Raw values; no smoothing.